

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MIDTERM EXAMINATION

MONTH & YEAR: APRIL 2024

Course: B. Tech

Semester: IV

Branch: CSE

Subject Name: Formal Language and Automata Theory

Subject Code: CSE 225

Duration: 90 minutes

Max Marks: 20

Instructions: Answer all Questions. Attempts all the parts of a question in continuation.

Roll No. _____

Name of the Student _____

(4)

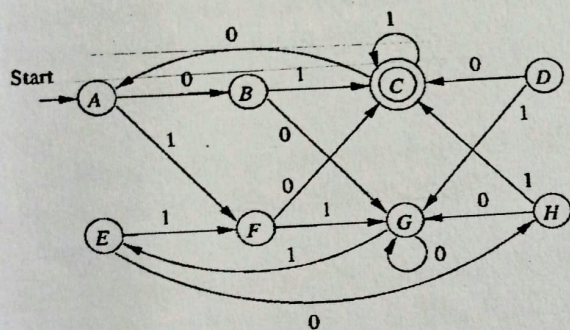
Find DFA for the following regular languages:

(a) All the strings of $\{0, 1\}$ in which, every substring of 3 symbols has at most two zeros.

(b) All the strings of $\{a, b\}$ do not end with substring 'aba'.

(4)

2. Minimize the following DFA.



(4)

3. Convert the following left linear regular grammar to equivalent right linear grammar

$$S \rightarrow Sab / Aa$$

$$A \rightarrow Bb / Aa / \epsilon$$

$$B \rightarrow Cba / d$$

$$C \rightarrow Ab / Ba / \epsilon$$

(4)

4. Design a mealy machine with 1-bit input and 2-bit output which behaves as follows:

Let z_k, n_k denote the no. of 0's and 1's respectively in initial k bits of the input ($z_k + n_k = k$). The machine O/Ps 00 until one of the following conditions holds.

- $z_k - n_k = 2$. In this case, the O/P at the k th and all subsequent clock ticks is 10.
- $n_k - z_k = 2$. In this case, the O/P at the k th and all subsequent clock ticks is 01.

(4)

5. Find the Context Free Grammar for the following languages:

(a) $L = \{a^n b^m c^k \mid n \geq k \text{ or } n \leq m \text{ where } n, m, k \geq 0\}$

(b) $L = \{n_a(w) = 2n_b(w) \mid w \in (a+b)^*\}$

	(b) (i) Construct a grammar G accepting the set L of all strings over $\{a, b\}$ having more a's than b's. (ii) Find the regular grammar for the following language $L = \{a^n b^m c^k \mid n > 1, m < 2, k > 3\}$	
4.	(a) For the language L, develop an appropriate Turing Machine. Include the transition function, state diagram, and justify the correctness of your design. $L = \{a^n b^{2n} \mid n \geq 0\}$ (b) Construct PDA for the following Language L $L = \{w_1 \# w_2 \mid w_1, w_2 \in (a+b)^*, \# \text{ is a terminal and } w_2 \text{ is 1's complement of reverse of } w_1\}$	5+5
5.	(a) Construct PDA for the following Context Free Grammar $G(\{S, A, B, C\}, \{a, b, d\}, S, P)$ the set of productions P of G are as follows: $S \rightarrow Sa / BC$ $A \rightarrow aA / b$ $B \rightarrow aB / \lambda$ $C \rightarrow aAb / bBa / d$ (b) Consider the following PDA $M = (\{q_0, q_1\}, \{a, b, c\}, \{a, c, z\}, \delta, q_0, \{q_1\})$ with the following transitions: $\begin{aligned} \delta(q_0, a, z) &= \{(q_0, az)\}, & \delta(q_0, a, a) &= \{(q_0, aa)\}, \\ \delta(q_0, a, c) &= \{(q_0, \lambda)\}, & \delta(q_0, b, z) &= \{(q_0, az)\}, \\ \delta(q_0, b, a) &= \{(q_0, aa)\}, & \delta(q_0, b, c) &= \{(q_0, \lambda)\}, \\ \delta(q_0, c, z) &= \{(q_0, cz)\}, & \delta(q_0, c, a) &= \{(q_0, \lambda)\}, \\ \delta(q_0, c, c) &= \{(q_0, cc)\}, & \delta(q_0, \lambda, z) &= \{(q_1, \lambda)\} \end{aligned}$ Where z indicate bottom marker of the stack. λ represents string end marker and pop configuration of the stack. Describe the language accepted by the above PDA by final state	5+5
6.	(a) Design a Turing Machine for given language $L = \{www \mid w \in \{a, b\}^+\}$ (b) Construct Context Sensitive Grammar for the following Languages. (i) $L = \{a^n b^m c^n d^m \mid n, m > 0\}$ (ii) $L = \{a^n b^n c^n \mid n > 1\}$	5+5